MakanMap

Crowd Level Prediction System

An End-To-End Machine Learning Pipeline
DSA Industry Project

PROBLEM & SOLUTION

THE PROBLEM

- ✕ Unpredictable crowd surges at food courts.
- ✕ Long queues & extended waiting times
- ✕ Vendors struggle with staffing & inventory
- ✕ No predictive tools for customers or operators

→

OUR SOLUTION

- ✓ ML-driven crowd level predictions
- ✓ Historical + contextual data integration
- ✓ Real-time REST API with <100ms latency
- ✓ Automated ETL pipelines via Apache Airflow

OBJECTIVES

<100ms

Inference Latency

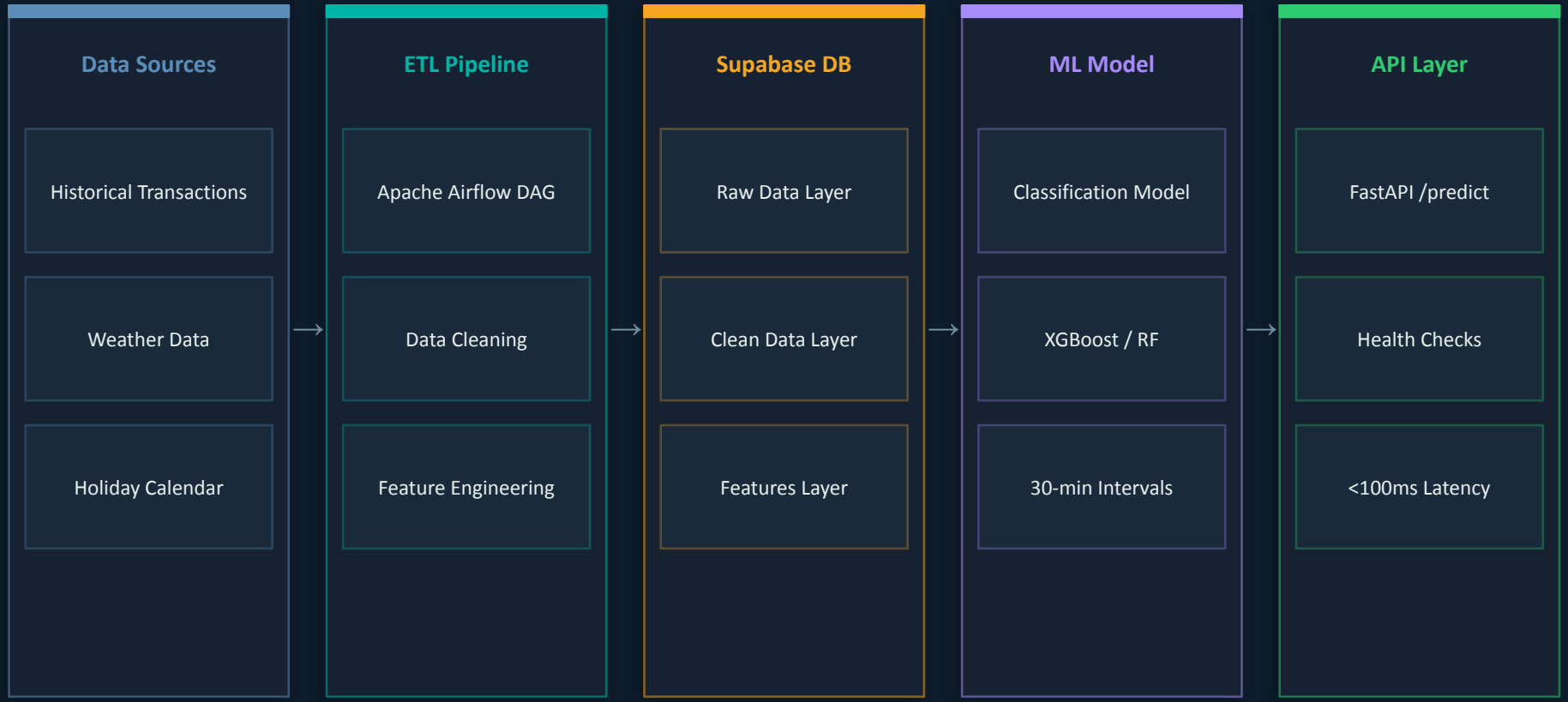
>85%

Prediction Accuracy

>99%

System Availability

SYSTEM ARCHITECTURE



DATA PIPELINE

01 Raw Data Layer

All ingested records stored as original format. No strict constraints — preserves dirty data for auditing.

- TEXT / FLOAT fields
- No data rejection
- Source of truth

→

02 Clean Data Layer

Processed after cleaning & validation. Proper types enforced. Business rules applied.

- Operating hrs: 6AM–10PM
- Celsius normalization
- Median imputation

→

03 Feature Layer

ML-ready features with engineered variables. 11 features, ~200K records processed in seconds.

- Cyclical time encoding
- Weather encoding
- Location popularity

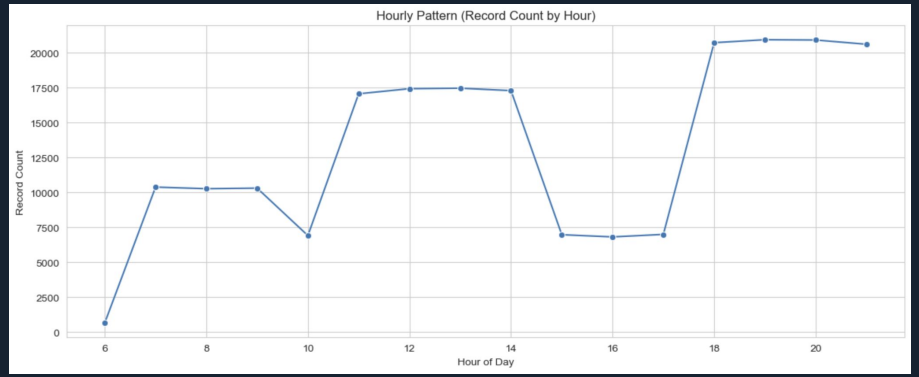
⚡ Orchestrated by Apache Airflow DAG | Manually triggered for production validation | ~200,000 records processed in seconds

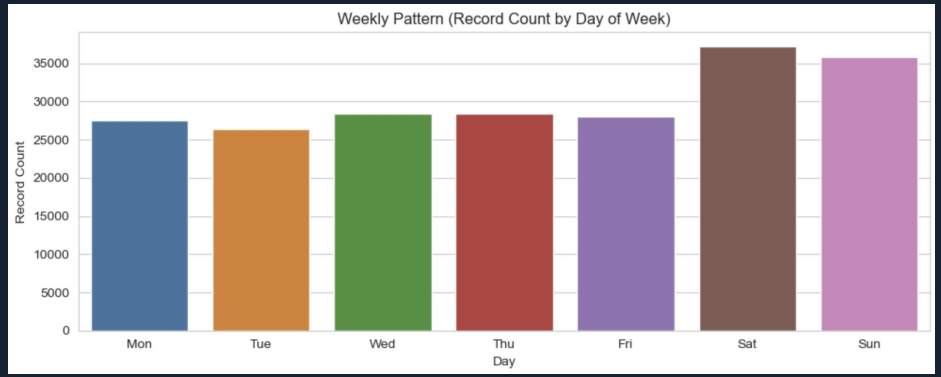
EXPLORATORY DATA ANALYSIS

01 Temporal Patterns

Time is the strongest predictor

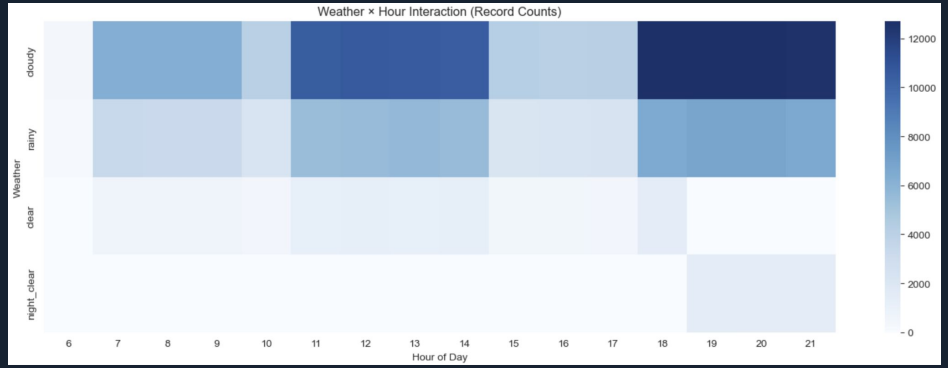
- Peak hours identified at 6PM
- Distinct Weekday vs Weekend behaviors



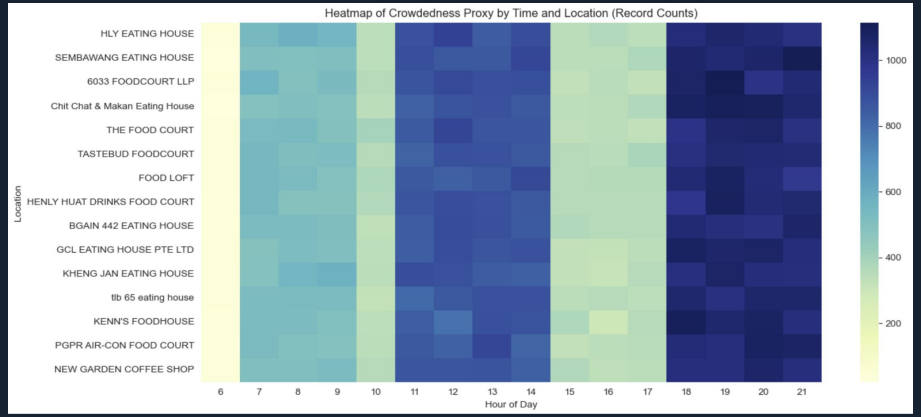


Beyond Time: Interaction Effects

- Weather + Time: Certain hours become much more crowded depending on the weather

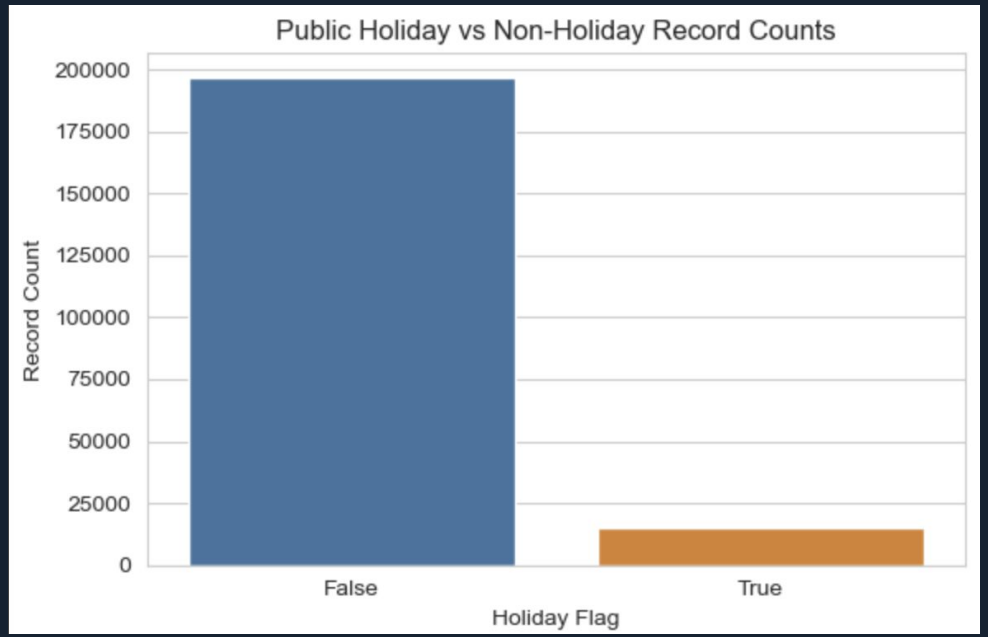


- Location + Time: Activity isn't just about "when," but "where" that time occurs



Beyond Time: Interaction Effects

- Holidays: Act as a disruptor to normal weekday routines



ML MODEL DEVELOPMENT

Two Approaches Explored

Experiment 1: Regression (30-min bins)

Model (Regressors)	MAE	RMSE	R^2
Gradient Boost (Production Model)	**0.0901**	**0.1189**	**0.239**
Random Forest	0.0970	0.1193	0.233
XGBoost	0.0861	0.1175	0.259

Experiment 2: Classification (30-min bins)

Target Crowd level: Low / Medium / High

Binning Quantile-based segmentation

Encoding One-hot for categorical variables

Imputation Median (numeric) + Most frequent (cat)

Training Setup

- Time-based 80/20 split
- Prevents data leakage
- Future data never predicts past
- Simulates real deployment
- Generalizes to unseen time periods

PREDICTION API

FastAPI REST Service

POST `/predict`

Predict crowd levels for next N time bins

GET `/health`

Docker health check — confirms model loaded

Input Parameters

- `temperature`
- `humidity`
- `weather`
- `location_frequency`
- `is_public_holiday`

API Response

```
timestamp:
2024-01-15 12:30:00
```

```
crowd_score:
0.72
```

```
label:
"High"
```

Key Design Decisions

- Model loaded once at startup
- Full sklearn pipeline in .pkl
- Consistent encoder at inference
- Auto time-bin derivation

FUTURE WORK

Queue Length Forecast

Use crowd level scores to estimate queue sizes

Automated Model Retraining

Use MLFlow to retrain the models on new data on a regular schedule

Wait-Time Predictions

Derive waiting time estimates from crowd levels

Real-Time Data Feeds

Live sensor/transaction data for streaming predictions

Q&A